

## TITLE PAGE

- **Problem Statement ID** – 26146
- **Problem Statement Title-** AI-Powered Monitoring and Analysis of Bitcoin Transaction Traffic
- **Theme-** Blockchain and Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name** - Blue Cipher

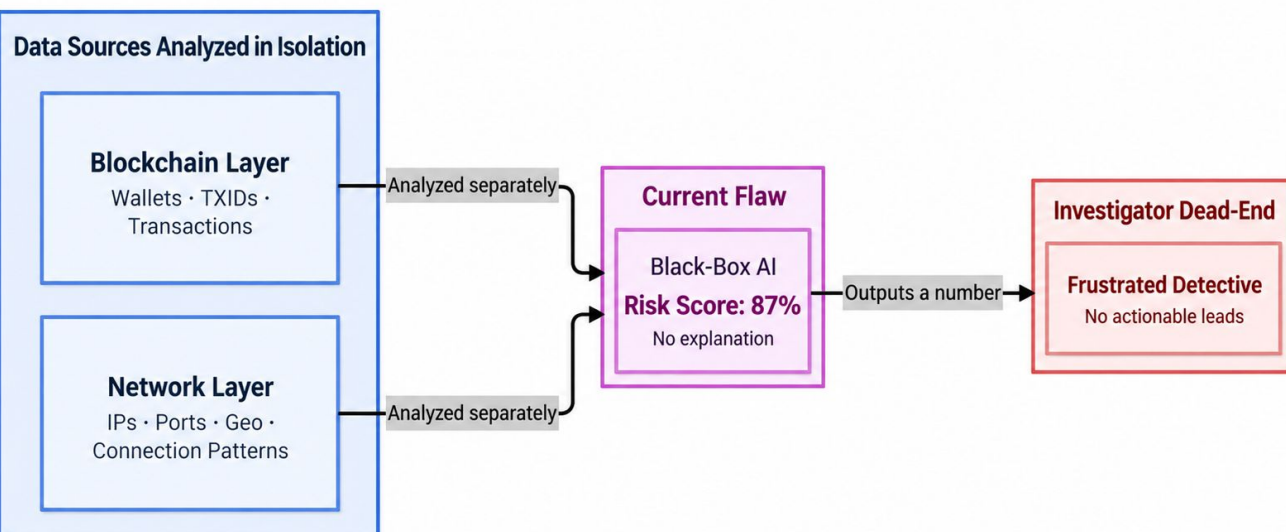


## Problem

- **Bitcoin's pseudonymous P2P design** enables illicit funds from ransomware, darknet and extortion to move and launder undetected.
- **Blockchain and network-layer signals are analyzed separately**, missing critical cross-layer correlations between wallets, transactions, IPs and timing.
- **Existing tools produce black-box risk scores from fragmented metadata**, leaving investigators without explainable, actionable leads.

## Proposed Solution

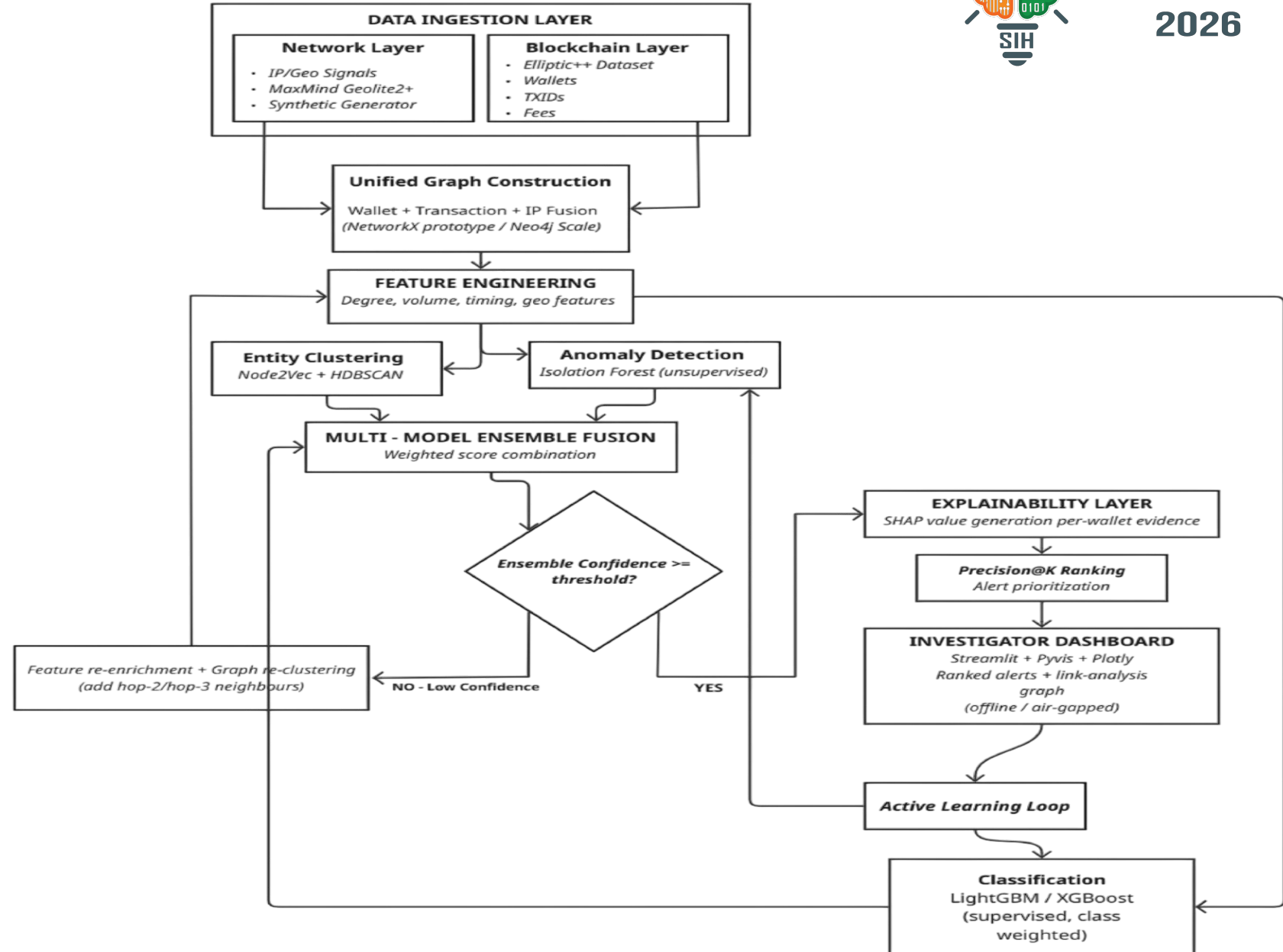
- **Fuse blockchain + network-layer signals::** Correlate wallets/transactions with IP/geo indicators from P2P node monitoring (entry-node timing/propagation), simulated via synthetic IP generator for this prototype, in one unified graph.
- **Multi-model AI ensemble:** Combine anomaly detection, supervised classification, and graph-based clustering for accurate, robust detection, even on Elliptic++'s highly imbalanced, mostly-unlabeled data (only 23% labeled; just 2% of the total dataset marked illicit).
- **Explainable by design:** Every flagged wallet comes with SHAP-based evidence, a human-readable, reason, not a black-box guess.
- **Ranked, offline-first dashboard:** Precision@K-ranked alerts paired with an interactive link-analysis graph, so investigators know exactly who to chase first, fully functional in secure, air-gapped environments.
- **Target Users:** Law Enforcement Agencies, Financial Intelligence Unit (FIU-IND), NTRO / Intelligence Agencies, Crypto Exchanges (VASPs)



## Tech Stack:

- **Data Processing:** Python, Pandas, Pydantic
- **Graph Engine:** NetworkX / Neo4j
- **Anomaly Detection:** Isolation Forest (Scikit-learn)
- **Classification:** LightGBM / XGBoost
- **Entity Clustering:** Node2Vec + HDBSCAN
- **Graph Neural Net (stretch):** GCN / PyTorch
- **Explainability:** SHAP
- **Dashboard:** Streamlit, Pyvis, Plotly
- **Data Sources:** Elliptic++, MaxMind GeoLite2, network-layer IP/timing signals simulated via synthetic generator for prototype
- **Deployment:** Offline / air-gapped
- **Testing/Validation:** Precision@K, F1-score on Elliptic++ labeled subset
- **Version Control:** Git/Github

Github Link : [<Link>](#)



## Risks & Mitigations

Graph scalability

Indexed storage + selective graph analysis

Class imbalance

Class weighting + Precision@K / Recall@K

Anonymized features

Feature curation + SHAP explanations

Limited network-layer  
identifiers

Synthetic IP/timing generator (prototype)

### Feasibility

- **Offline-first:** No internet/live node needed
- **Open-access dataset:** Elliptic++ (203K transactions, 822K wallets), freely available research dataset, no licensing cost
- **Proven ML tools:** Established models, open-source libraries
- **Modular design:** Stages built/tested independently

### Commercial Viability

- **Target users:** Cybercrime units, FIUs, exchanges, investigators
- **Growing need:** Rising crypto-crime + AML focus
- **Low deployment cost:** Mainly open-source
- **Future scope:** Adaptable to other blockchain networks

| Impact Area                | Existing Solutions                     | Example                                                                | ChainSentry                                       |
|----------------------------|----------------------------------------|------------------------------------------------------------------------|---------------------------------------------------|
| Explainability             | Enterprise-grade, but proprietary/paid | Elliptic's evidence trails are built into paid tiers only              | Open-source SHAP, fully transparent methodology   |
| Cost/Access                | Expensive licensing, enterprise-only   | Chainalysis: \$50K–\$200K/yr; TRM Labs: \$50K–\$500K/yr, no free tier  | <b>Free</b> , open-source stack                   |
| Deployment                 | Cloud/SaaS-based                       | Chainalysis Reactor, Elliptic Nexus — cloud platforms                  | Fully <b>offline</b> / air-gapped by design       |
| Network+Block chain fusion | Primarily on-chain focused             | Chainalysis traces 27+ chains via swaps/bridges, but focus is on-chain | Explicit graph fusion of wallet+IP+network layers |

Sources: CostBench pricing benchmarks, vendor product pages (2026)

Explainable investigation. Offline by design. Open by nature.

### StakeHolder Benefits

- **Investigators:** Ranked, evidence-backed leads with less manual triage
- **Exchanges & FIUs:** Faster AML investigation and suspicious-activity review
- **Law Enforcement:** Secure, offline-first deployment for sensitive cases
- **Regulators:** Transparent and auditable detection logic

### Future

- **Multi-chain expansion:** Extend beyond Bitcoin to Ethereum, stablecoins, other traceable ledgers
- **Scalable investigations:** Support larger transaction graphs and datasets
- **Real network data:** Replace synthetic IPs with legally-sourced data via ISP/law-enforcement partnerships
- Advanced behavioral detection: Expand peeling-chain, CoinJoin and burst-pattern analysis
- **Cross-domain use:** Apply graph + explainability framework to insurance, banking, supply-chain fraud

## 1. Problem Validation

Parliamentary Standing Committee on Home Affairs (2025): Cyber Crime - Ramifications, Protection and Prevention, 53.9 lakh cybercrime complaints; ~₹31,594 crore reported up to Nov. 2024.

[\[Report\]](#)

b) Chainalysis (2026): 2026 Crypto Crime Report, \$154B received by illicit addresses in 2025 (+162% YoY)

[\[Report\]](#)

## 2. Technical Feasibility

a) Elmougy & Liu (2023): *Elliptic++: An Enhanced Dataset for Bitcoin Address and Transaction Analysis*, *elliptic++ dataset for illicit transaction and entity analysis*.

[\[Paper\]](#)

b) Lundberg & Lee (2017): *A Unified Approach to Interpreting Model Predictions— Model Predictions, SHAP framework for explainable AI*.

[\[Paper\]](#)

## 3. Market & Application Validation

a) TRM Labs (2026): 2026 Crypto Crime Report, \$158B incoming value to illicit entities in 2025, highlighting the need for crypto-crime investigation tools.

[\[Report\]](#)